



BatchIt

Buy Together, Save Together

Mobile + REST API Application

Flutter (Dart) • Django REST Framework • PostgreSQL

FLUTTER

DJANGO

POSTGRESQL

JENKINS

Introduction & Problem Statement

What is BatchIt?

A **mobile-first collaborative bulk purchasing** platform. Consumers form *buying groups (batches)* to unlock wholesale prices on essential goods.

Customers — search & join batches

Providers — verified businesses listing products

Admins — verify providers, ensure trust

Inspired By

Groupon (2008) model + Pinduoduo (2015) adapted for **mobile-first Sub-Saharan African commodity markets**.

The Problem

High per-unit retail prices for individuals

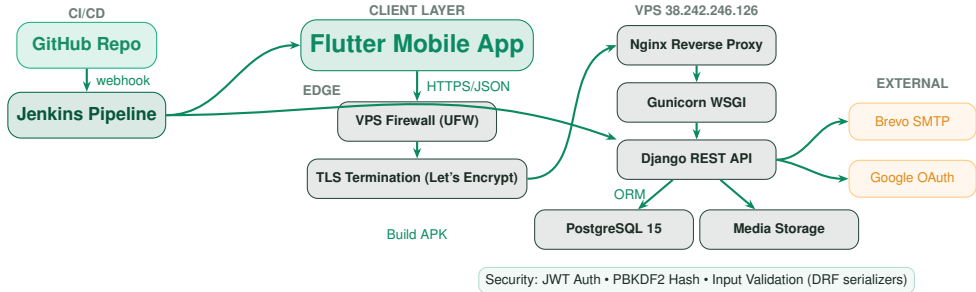
No digital channel for group commitment purchases

Limited access to wholesale channels

No verification/trust for small vendors

BatchIt solves this

System Architecture



Key Features

Batch Lifecycle Management

Create → Discover → Join → Fill → Complete. Live **progress bar**. Auto status transitions (open/filled/expired).

Dual Authentication

Email/password + **6-digit code** via Brevo SMTP, plus **Google OAuth 2.0**. JWT: 60 min access, 7 day refresh.

Provider Verification Workflow

Business submits docs → Admin reviews → Approved/Rejected with in-app notification.

Batch Group Chat

Each batch has a **persistent group chat room**. REST polling (POST messages / GET history). Coordinate with buyers.

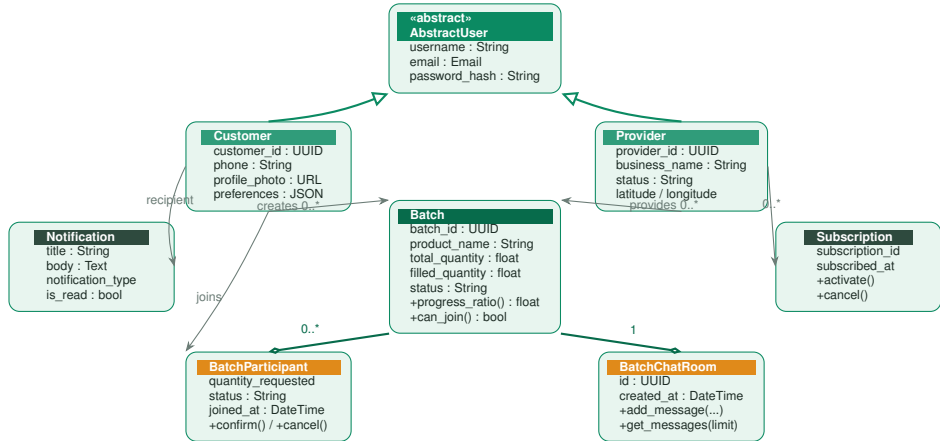
Map-Based Discovery

OpenStreetMap via `flutter_map`. View providers and batches by **geographic proximity**. Tap pins for detail.

In-App Notifications

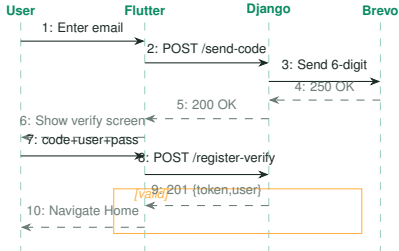
Triggered on batch creation, join, fill, and provider approval. Unread badge count. Provider subscription model.

Object-Oriented Design — Core Class Diagram

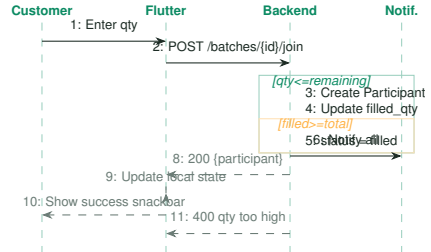


Critical Sequence Flows

SD1 — Email Registration + Verification



SD4 — Customer Joins a Batch



Scrum Methodology & Team Organisation

4 Sprints • GitHub Projects Kanban • Feature Branches

S1 Auth + CI/CD Setup

S2 Batch Core + Join Logic

S3 Provider + Chat + Notifications

S4 OAuth + 315 Tests + Deploy

Must-Have: 68 pts • Should-Have: 18 pts • Could-Have: 5 pts

GitHub • WhatsApp standups • VS Code • Android Studio

Team Roles (5 Members)



Kamdeu Yamdjeuson
Scrum Master / DevOps & VPS Setup



Njume Leslie
Product Owner / Backend Lead



Atsimbong Joshua
Flutter Developer / UI & Maps



Chijioke Emmanuel
Backend Developer / Realtime & Chat



Fru Chi
Flutter Developer / Core Features

Conflicts resolved: pubspec.yaml ownership;
Django migration squash per sprint

Core Algorithms & API Design

Batch Fill Algorithm (Django views.py)

```
1  def join_batch(request, batch_id):
2      batch = Batch.objects.get(batch_id=batch_id)
3      qty = request.data['quantity_requested']
4
5      # Guard: batch must be open
6      if batch.status != 'open':
7          raise ValidationError("Batch is not open.")
8
9      # Guard: quantity check
10     remaining = (batch.total_quantity
11                 - batch.filled_quantity)
12
13     if qty > remaining:
14         raise ValidationError(
15             f"Only {remaining} units left.")
```

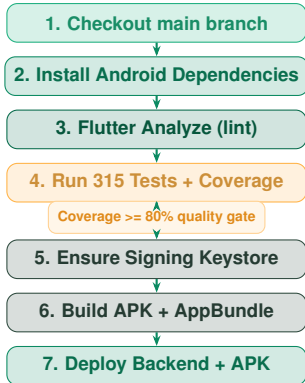
Selected REST Endpoints

POST	/auth/login/
POST	/auth/register-verify/
GET	/batches/?status=open
POST	/batches/{id}/join/
POST	/providers/register/
POST	/admin/providers/{id}/verify/
GET	/batches/{id}/chat/messages/
POST	/batches/{id}/chat/messages/

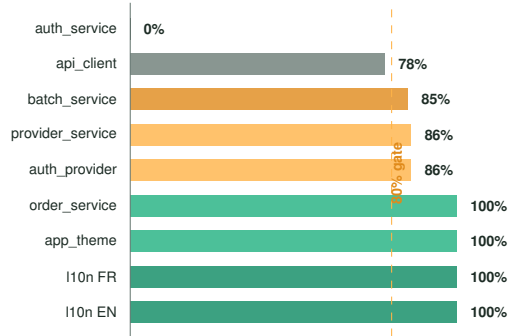
40+ endpoints • JWT Auth
Swagger/OpenAPI via drf-yasg
PBKDF2 password hashing

CI/CD Pipeline & Test Coverage

Jenkins Pipeline — Triggered on
Every Push to main



Test Coverage — Build #46 (83.6% overall)



315 tests — all passing
83.6% line coverage (1466 / 1753 lines)
Packages: mocktail + flutter_test
Started at 28.2% — improved systematically

Results, Future Work & Conclusion

Achievements

40+
API Endpoints

315
Tests Passing

83.6%
Coverage

4 Sprints
Scrum delivery

14 min
CI/CD pipeline

- Full-stack Flutter + Django in 4 weeks
- Bilingual EN/FR Material Design 3 UI
- Production VPS: `batchit.duckdns.org`
- Signed APK auto-built via Jenkins
- 40+ documented REST API endpoints

Challenges Overcome

Jenkins keystore path • Google SHA-1 mismatch
Coverage 28% → 83.6% • JWT migration

Recommendations for Future Work

- WS** Real-time chat with Django Channels + Redis
- FCM** Firebase push notifications (Android + iOS)
- PAY** MTN MoMo / Orange Money payment integration
- GEO** PostGIS spatial filtering by GPS radius
- iOS** iOS App Store deployment + Let's Encrypt HTTPS
- WEB** React provider dashboard (web admin panel)

BatchIt — Buy Together, Save

SEN2241 • Group 13 • ICT University • Spring 2026

Instructor: Tekoh Palma Achu GitHub:

github.com/Kynmmarshall/BatchIt Live: `batchit.duckdns.org`